THE FIVE COMPARISON PROPERTIES HOLD FOR THE H-TOPOLOGY OVER THE SMALL ETALE SITE WITH PROPER-MORPHISM COEFFICIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the weak Serre verification with the exactness argument for the extension condition, and the five comparison properties with the reason and reference for each.

Editorial note. The label lemma-compare-h-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-compare-h-etale –start 22647 –end 22715 returned RESULT=OK (occupancy FREE).

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Cohomological descent for the etale comparison. Same chapter, the lemma stating that K -> Ra_{X,*}a_X^{-1}K is an isomorphism for torsion coefficients, which is used for the exactness and fullness of pi_X^{-1}. Supplies the weak Serre verification and the identification G = pi_X^{-1}pi_{X,*}G.
- The description of the pullback along the comparison. Same chapter, the lemma identifying pi_X^{-1} of a torsion sheaf with the rule given by the sections of its small pullback. Supplies property (item-A-sheaf) and the shape of the objects of A'_X.
- Base change and torsion for proper morphisms. Morphisms, the permanence of properness and of finite presentation under base change; Same chapter, the comparison lemma for proper morphisms and the lemma that pushforwards of torsion sheaves are torsion. Supplies the base change property and property (item-A-and-P).
- Domination of an fppf cover by a finite locally free one. More on Morphisms, the lemma producing a finite locally free cover dominating an fppf cover, and More on Flatness, the lemma that a surjective proper finitely presented morphism is an h-covering. Supplies property (item-refine-tau-by-P).

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0ETW: sources/stacks-chow/tags.txt, source lines 18531–18531. Prerequisite pointer invoked at proof line(s) 22714; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Prerequisite pointer invoked at proof line(s) 22669; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDI: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21308–21308. Prerequisite pointer invoked at proof line(s) 22671; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21323–21323. Prerequisite pointer invoked at proof line(s) 22702; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EW8: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22554–22554. Prerequisite pointer invoked at proof line(s) 22698; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDD: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 13924–13924. Prerequisite pointer invoked at proof line(s) 22705; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DET: sources/stacks-chow/tags.txt, source lines 16813–16813. Prerequisite pointer invoked at proof line(s) 22710; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01TS: sources/stacks-chow/tags.txt, source lines 2385–2385. Prerequisite pointer invoked at proof line(s) 22689; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Prerequisite pointer invoked at proof line(s) 22686; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ7: sources/stacks-chow/tags.txt, source lines 18717–18717. Prerequisite pointer invoked at proof line(s) 22660; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ6: sources/stacks-chow/tags.txt, source lines 18716–18716. Prerequisite pointer invoked at proof line(s) 22659; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ4: sources/stacks-chow/tags.txt, source lines 18714–18714. Prerequisite pointer invoked at proof line(s) 22657; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ8: sources/stacks-chow/tags.txt, source lines 18718–18718. Prerequisite pointer invoked at proof line(s) 22661; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ5: sources/stacks-chow/tags.txt, source lines 18715–18715. Prerequisite pointer invoked at proof line(s) 22658; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZ3: sources/stacks-chow/tags.txt, source lines 18713–18713. Prerequisite pointer invoked at proof line(s) 22663; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 021I: sources/stacks-chow/tags.txt, source lines 2671–2671. Prerequisite pointer invoked at proof line(s) 22694; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider the comparison morphism $\epsilon : (Sch/S)_h \to (Sch/S)_{étale}$. Let $\mathcal{P}$ denote the class of proper morphisms. For X in $(Sch/S)_{étale}$ denote $\mathcal{A}'_X \subset Ab((Sch/X)_{étale})$ the full subcategory consisting of sheaves of the form $\pi_X^{-1}\mathcal{F}$ where $\mathcal{F}$ is a torsion abelian sheaf on $X_{étale}$ Then Cohomology on Sites, Properties (0EZ4), (0EZ5), (0EZ6), (0EZ7), and (0EZ8) of Cohomology on Sites, Situation 0EZ3 hold.*

Proof. We first show that $\mathcal{A}'_X \subset Ab((Sch/X)_{étale})$ is a weak Serre subcategory by checking conditions (1), (2), (3), and (4) of Homology, Lemma 0754. Parts (1), (2), (3) are immediate as π_X^{-1} is exact and fully faithful for example by Lemma 0DDI. If $0 \to \pi_X^{-1}\mathcal{F} \to \mathcal{G} \to \pi_X^{-1}\mathcal{F}' \to 0$ is a short exact sequence in $Ab((Sch/X)_{étale})$ then $0 \to \mathcal{F} \to \pi_{X,*}\mathcal{G} \to \mathcal{F}' \to 0$ is exact by Lemma 0DDI. In particular we see that $\pi_{X,*}\mathcal{G}$ is an abelian torsion sheaf on $X_{étale}$. Hence $\mathcal{G} = \pi_X^{-1}\pi_{X,*}\mathcal{G}$ is in $\mathcal{A}'_X$ which checks the final condition.

Cohomology on Sites, Property (0EZ4) holds by the existence of fibre products of schemes, the fact that the base change of a proper morphism of schemes is a proper morphism of schemes, see Morphisms, Lemma 01W4, and the fact that the base change of a morphism of finite presentation is a morphism of finite presentation, see Morphisms, Lemma 01TS.

Cohomology on Sites, Property (0EZ5) follows from the commutative diagram (3) in Topologies, Lemma 021I.

Cohomology on Sites, Property (0EZ6) is Lemma 0EW8.

Cohomology on Sites, Property (0EZ7) holds by Lemma 0DDJ part (2) and the fact that $R^i f_{small}\mathcal{F}$ is torsion if $\mathcal{F}$ is an abelian torsion sheaf on $X_{étale}$, see Lemma 0DDD.

Cohomology on Sites, Property (0EZ8) is implied by More on Morphisms, Lemma 0DET combined with the fact that a surjective finite locally free morphism is surjective, proper, and of finite presentation and hence defines a h-covering by More on Flatness, Lemma 0ETW. □